

Yuguang Li

yuguang.lee1@gmail.com | (315) 897-7889 | Seattle (Open to relocate) | [Website / Research Portfolio](#) | [Patent Portfolio](#)

Technical Profile: Principal Applied Scientist and computer vision researcher building production-scale multi-view perception and 3D reconstruction systems. My work combines multi-view geometric reasoning, multi-modal visual reasoning, vision foundation models, geometric optimization, generative models, and deployable ML infrastructure. At Zillow, I led the technical development of the company's first production floor-plan reconstruction platform, scaling sparse wide-baseline indoor reconstruction from research prototypes into deployed workflows supporting hundreds of thousands of floor-plan-centric home experiences annually. Research output includes a first-author CVPR 2025 Highlight, a first-author NeurIPS 2026 submission, several CVPR/ECCV publications with interns, large-scale datasets, deployed ML systems, and 30+ patents/patent filings.

Core Areas: Multi-view perception | 3D reconstruction | Camera pose estimation | Bundle adjustment / geometric optimization | Structured indoor scene understanding | Vision foundation models | Generative models / diffusion | Dataset annotation and evaluation design | Production ML systems | Graphics / image-based rendering | Geometric shape processing

EXPERIENCE

Zillow Group

Aug. 2017 – Present

Principal Applied Scientist / Technical Lead

Seattle

Continuously role since 2017: full-time 2017-2023 • Zillow-sponsored part-time PhD research 2024-2026 • returning full-time 2026

Focus: Production-scale indoor reconstruction from sparse, unposed, wide-baseline panoramas

Zillow Floor-plan Reconstruction Platform Originator • Technical Lead • System Architect (2018-2020)

- **Created demand for production-scale floor-plan media:** I originated the floor-plan-centered consumer experience, led the first-place Zillow Hack Week project, and patented the interaction concept that became the conceptual starting point for Zillow's [Showcase](#) experience. In parallel, I proved the reconstruction workflow with an annotation pilot program, helping secure executive buy-in for broader investment in scalable floor-plan reconstruction; the resulting floor-plan media experience now supports hundreds of thousands of home experiences annually.
- **Originated the reconstruction platform:** I defined Zillow's floor-plan reconstruction platform as a sparse multi-view indoor reconstruction system, defining the spatial representations, data contracts, reconstruction stages, and interactive geometry operations that still underlie the platform today. The system spans image geometric normalization, global metric-scale calibration, single-view layout estimation, camera-pose recovery/re-localization, multi-view geometry merging, and geometrically consistent floor-plan output. This platform later became the basis for the [ZInD dataset/system paper](#).
- **Built the precision reconstruction core:** I owned the design and prototype of the core CV/graphics system for localization, pose refinement, incremental room merging, and floor-plan geometry cleanup. This interactive reconstruction system made centimeter-level floor-plan production repeatable from sparse panoramas and became a key technical reason the workflow scaled to hundreds of thousands of homes per year before full automation.
- **Architected for modular ML automation:** I designed the platform around decomposable geometric representations and separable reconstruction stages, allowing learned components to replace targeted annotation and QA steps while preserving the final floor-plan geometry contract and centimeter-level production accuracy.

Next-generation Autonomous Multi-view Reconstruction First-author Research Lead *BADGR* • *MvFFN/MOIS* (2024-2026)

- **Led a Zillow-backed autonomous reconstruction agenda:** Building on the production impact of the floor-plan platform I originated, I received leadership support to pursue autonomous floor-plan reconstruction research at PhD depth, with broad technical freedom and collaboration across top-tier UW and industry researchers. The research focuses on jointly recovering camera poses, wall geometry, and centimeter-level multi-room floor plans from sparse, unposed, wide-baseline panoramas.
- **First-authored [BADGR](#), CVPR 2025 Highlight:** *BADGR* is a learned global refinement framework for centimeter-accurate floor-plan reconstruction from sparse wide-baseline images. It conditions a floor-plan-specific *DDPM* diffusion model on Levenberg–Marquardt-style layout-pose bundle-adjustment gradients, enabling the model to refine camera poses and wall geometry while learning outlier rejection and layout inpainting for noisy or occluded regions. I originated the idea, implemented the full method, ran all experiments, and led the paper writing.
- **First-authored [MvFFN/MOIS](#), NeurIPS 2026 submission:** I developed the first learned end-to-end system for multi-view multi-room floor-plan reconstruction from sparse, unposed, wide-baseline panoramas, without assuming known camera poses. After direct floor-plan prediction proved brittle under limited and noisy data, I reformulated the learning problem through our novel *MOIS* formulation and built *MvFFN* as a multi-view geometry reasoning model. *MvFFN* combines geometric foundation-model features, semantic/geometric cross-view reasoning, dense segmen-

tation/geometric losses, and multi-level inductive biases from pairwise cues to stabilize learning from masked, scarce, and inconsistently annotated data while recovering pixel-grounded cross-view wall identities, ordered wall structure, and within-room connectivity for downstream floor-plan assembly.

- **Built the data foundation for autonomous reconstruction:** I created the **ZInD-CrossView** dataset from sparse, visually overlapping indoor panorama sets with recovered camera poses, globally consistent wall-instance IDs, global room layouts, and densely rendered cross-view segmentation labels, enabling the *MvFFN/MOIS* work. I also assembled roughly 60k semantic floor plans as the **FloorPlan-60K** dataset to train *BADGR* diffusion model, providing large-scale layout examples for learning generative floor-plan structural priors.
- **Driving productization back in Zillow:** I recently started leading the productionization efforts for autonomous floor-plan reconstruction as I return to full-time technical leadership, defining the next floor-plan automation strategy around *BADGR/MvFFN* and their integration into Zillow's production reconstruction pipeline.

Production ML Automation for Human-in-the-loop Reconstruction (2021-2023)

- **Owned the floor-plan ML automation agenda:** As the principal scientist driving floor-plan automation, I defined the roadmap across pose, layout, relocalization, and geometry cleanup; estimated module-level cost savings; and tracked progress from UI-action counts to end-to-end annotation time. The program reduced floor-plan production from roughly 50 minutes in the manual proof-of-concept to roughly 15 minutes in the ML-assisted workflow.
- **Defined and mentored pose automation research:** I identified wide-baseline indoor pose estimation as a core automation problem and initiated research efforts for featureless scenes, extreme viewpoint changes, and sparse panorama overlap. As second author and senior technical mentor on both ECCV 2022 papers, I shaped the directions behind *SALVe*, a birds'-eye view (BEV) / semantic pose-estimation approach with pose-graph optimization, and *CovisPose*, a robust transformer model for wide- and extreme-baseline panorama pose estimation using learned indoor structural correspondences, with a model design that resembles *DUST3R*.
- **Productized *CovisPose* end-to-end:** I led *CovisPose* from research model to production localization system, owning the technical direction, deployment metrics, operator-feedback loop, and geometric post-processing design while coordinating cross-functional execution on model scaling, data expansion/re-annotation, system integration, and UI iteration. The deployed system improved primary panorama selection and coarse floor-plan regeneration, reducing floor-plan annotation time by roughly 30%.
- **Originated *PSM-Net* for multi-view layout reconstruction:** I originated the BEV-based multi-view layout estimation approach behind *PSM-Net*, built the initial proof of concept into a trainable model, then recruited and mentored an intern and junior scientist to mature the method into a second-author CVPR 2022 publication.
- **Improved single-view layout automation:** I co-led the production iteration of the monocular room-layout model with a junior scientist, improving the model through stronger backbones, loss design, uncertainty estimation, and data hygiene. These changes reduced manual correction and contributed to roughly 25% reduction in layout-related annotation time.

Image-based Rendering and Virtual Touring Technical Lead • Graphics Prototype

- **Built low-cost virtual touring graphics:** I designed an OpenGL-shader-based rendering approach, later patented with me as first-named inventor, for immersive indoor transitions between sparse, wide-baseline panoramas using floor-plan geometry and predicted depth, avoiding the need for depth sensors or dense high-fidelity 3D reconstruction. I also led cross-functional integration with front-end and full-stack engineers.

Leadership / IP: Principal scientist and technical lead across floor-plan reconstruction, pose estimation, layout estimation, autonomous reconstruction, and visual AI; senior mentor on CVPR/ECCV publications with interns and junior scientists; inventor on 30+ patents/patent filings covering floor-plan reconstruction, virtual touring, camera pose estimation, image-based rendering, and AI-assisted media.

University of Washington – Zillow-sponsored PhD Research Collaboration

Jan. 2024 – Jun. 2026

Full-time PhD Researcher / Computer Vision & AI

Seattle

- **Zillow-sponsored research track:** after originating and scaling Zillow's floor-plan reconstruction platform, I received Zillow leadership support to pursue the full autonomous reconstruction problem at academic depth while remaining tied to Zillow's long-term production roadmap. PhD defense scheduled for June 2026.
- **Cross-industry and UW collaboration:** collaborated with UW researchers, PhD students, and senior industry researchers across Zillow, Apple, NVIDIA, and Microsoft on *BADGR* and *MvFFN/MOIS*; contributed research direction, mentoring, system formulation, and technical integration.

Adobe Creative Technology Labs

June 2016 – Sept. 2016

Researcher

Seattle

- Prototyped a high-performance cross-platform photo processing pipeline based on Halide compiler. Our work was later adopted as the computational photography framework in products like Photoshop, Lightroom, etc.

EDUCATION

PhD, Computer Vision/Artificial Intelligence, University of Washington

2024 - June 2026

MS, Computer Vision, University of Washington

MS, Geospatial Information Science, State University of New York

BE, Opto-electrical engineering, Beijing University of Aeronautics and Astronautics (top-5 in China)

TECHNICAL SKILLS

ML / Vision: PyTorch, model training/evaluation, diffusion models, multimodal foundation models, segmentation, pose estimation, bundle adjustment, geometric optimization.

Systems / Productization: Python, C++, AWS-based infrastructure, production ML deployment, data/annotation pipelines, experiment tracking, metric design.

Graphics / UI: OpenGL/GLSL shaders, image-based rendering, React, front-end prototyping.